SELMER ELLIPTIC CURVES

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The problems below study a single family of plane cubics and use it to reconstruct, by hand, some of the main results of [Sil09, §§III.1–III.4]: Weierstrass equations, the group law, the Riemann–Roch construction of a Weierstrass model, and a first endomorphism. Section 4 is a set of worked `Magma` examples to read and re-run afterwards, with three short variations to try at the end. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A field k with $\operatorname{char} k \neq 3$, an algebraic closure $\overline{k}$ of k, and $d \in k^\times$.
- The cubic form $F := X^3 + Y^3 - dZ^3$, the plane cubic $C_d := \{F = 0\} \subset \mathbb{P}^2_k$, and its rational point $O := (1 : -1 : 0)$.
- A primitive cube root of unity $\zeta \in \overline{k}$.
- A Selmer elliptic curve: a pair (C_d, O) of this shape, after [Sel51]. Section 5 explains the name.
- Projective coordinates X, Y, Z, as usual, and lowercase letters for affine coordinates.
- The base change $\overline{C}_d := C_d \times_k \overline{k}$.

How to use this. Sections 1–2 are meant for the problem session. Section 3 may be started there and finished at home. As a rough guide, allow about 12 minutes for §1 and 33 for §2. Within §2, Problem 2.2 is the core and Problem 2.5(5)–(7) the payoff. If time is short, leave Problem 2.5(1)–(4) and Problem 2.8 for later. Section 4 needs no time in the session. It is worked examples, not exercises.

1. Warm-up: a plane cubic and a rational point on it

Definition 1.1 (Elliptic curves [Sil09, §III.3])**.** An elliptic curve over k is a pair (E, O) with E a smooth, projective, geometrically irreducible curve of genus one over k and $O \in E(k)$ a fixed rational point.

Problem 1.2 (It is an elliptic curve)**.**
(1) Check that $O \in C_d(k)$.
(2) Compute the three partial derivatives of F and deduce that C_d is smooth, using $\operatorname{char} k \neq 3$ and $d \neq 0$. Where is each hypothesis used?
(3) Suppose instead that $\operatorname{char} k = 3$ (only for this problem). Show that over $\overline{k}$ the curve C_d is a *triple line*.[a]
(4) Show that a smooth plane curve is geometrically irreducible.[b]
(5) Conclude from the degree–genus formula that (C_d, O) is an elliptic curve over k.

[a] *Hint:* Frobenius.
[b] *Hint:* Bézout.

The group law in Problem 1.4(4) is the chord–tangent construction of [Sil09, §III.2].

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Proposition 1.3 (The group law and Pic^0 [Sil09, Proposition III.3.4])**.** *The map $P \mapsto [P - O]$ identifies the chord–tangent group law on $C_d(\overline{k})$ with addition in $\mathrm{Pic}^0 \overline{C}_d$.*

Problem 1.4 (The base point is an inflection point)**.**
(1) Show that the line $Z = 0$ meets C_d in three *distinct* points of $C_d(\overline{k})$, and write them down. For which k are all three in $C_d(k)$?
(2) Compute the tangent line L to C_d at O, and show that $L \cap C_d = 3O$. (So O is an inflection point, or flex.)
(3) In a Weierstrass model, which line plays the role of L? What is different here?
(4) Let $\oplus$ denote the group law on C_d with identity O, defined by the chord–tangent construction. Using (2), show that for $P, Q, R \in C_d(\overline{k})$,

$$P \oplus Q \oplus R = O \iff P + Q + R \text{ is cut out on } C_d \text{ by a line.}$$

2. The group law

Problem 2.1 (Negation, and 2-torsion)**.**
(1) Show that $\ominus(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$ for every point $(X_0 : Y_0 : Z_0) \in C_d(\overline{k})$. There are two things to check: that the right-hand side lies on C_d, and that it is the third intersection point of C_d with the line through O and $(X_0 : Y_0 : Z_0)$.
(2) Deduce that the nontrivial 2-torsion of C_d is cut out by the line $X = Y$, and that $\#C_d[2](\overline{k}) = 4$ when $\operatorname{char} k \neq 2$.
(3) Take $k = \mathbb{Q}$. For which d does C_d have a nontrivial rational 2-torsion point? Write it down, and give the smallest positive integer d for which this happens.

Part (2) of Problem 2.2 is its heart. Parts (4) and (5) are then mechanical. If time is short, take (2) on faith and come back to its proof, but do not skip (3): it is what makes (5) painless. The identity of (2) is the polarization of a cubic form. For a diagonal cubic the trilinear form can be written down by hand.

Problem 2.2 (The trilinear form, and the addition formulas)**.**
For vectors $P = (X_P, Y_P, Z_P)$, Q and R in $\overline{k}^3$, define

$$\langle P, Q, R\rangle := X_P X_Q X_R + Y_P Y_Q Y_R - d\, Z_P Z_Q Z_R. \tag{2.1}$$

(1) Check that (2.1) is symmetric in P, Q, R, linear in each argument separately, and satisfies $\langle P, P, P\rangle = F(P)$.
(2) Prove the *expansion identity*: for scalars s, t,

$$F(sP + tQ) = s^3\langle P, P, P\rangle + 3s^2 t\,\langle P, P, Q\rangle + 3st^2\,\langle P, Q, Q\rangle + t^3\langle Q, Q, Q\rangle. \tag{2.2}$$

Where do the 3's come from?[a]
(3) Prove that $3\langle P, P, Q\rangle = \nabla F(P) \cdot Q$, and deduce that for $P \in C_d(\overline{k})$,

$$\langle P, P, Q\rangle = 0 \iff Q \text{ lies on the tangent line to } C_d \text{ at } P.$$

(4) Let $P_1 = (X_1 : Y_1 : Z_1)$ and $P_2 = (X_2 : Y_2 : Z_2)$ be distinct points of $C_d(\overline{k})$, and parametrize the line through them as $sP_1 + tP_2$. Using (2), show that the third intersection point is at $(s : t) = (B : -A)$ where $A = \langle P_1, P_1, P_2\rangle$ and $B = \langle P_1, P_2, P_2\rangle$, and hence that

$$P_1 \oplus P_2 = (BY_1 - AY_2 : BX_1 - AX_2 : BZ_1 - AZ_2). \tag{2.3}$$

Why can A and B not both vanish?
(5) Let $P = (X_0 : Y_0 : Z_0) \in C_d(\overline{k})$ and put $Q := (Y_0^2 : -X_0^2 : 0)$. Verify that Q lies on the tangent to C_d at P, and use (2) and (3) to derive

$$[2]P = \big(-Y_0(2X_0^3 + Y_0^3) : X_0(X_0^3 + 2Y_0^3) : Z_0(X_0^3 - Y_0^3)\big). \tag{2.4}$$

(6) Check (2.4) against Problem 2.1: what does it give on the line $X = Y$? (The line $X = 0$ is taken up in Problem 2.5.)
(7) Where does the whole method break down if $\operatorname{char} k = 3$? Compare your answer with Problem 1.2(3).

[a] *Hint:* expand $\langle sP + tQ, sP + tQ, sP + tQ\rangle$ by trilinearity into eight terms and collect using symmetry.

Problem 2.3 (Arithmetic on C_7)**.**
Let $d = 7$, so that $P := (2 : -1 : 1) \in C_7(\mathbb{Q})$ because $2^3 + (-1)^3 = 7$.
(1) Compute $[2]P$ using (2.4), and simplify. Verify your answer by exhibiting an identity between cubes of integers.
(2) Compute $[3]P$ using (2.3) applied to P and $[2]P$, and verify it the same way.
(3) Compute $[4]P$.
(4) The three rational points you have found have numerators and denominators of rapidly increasing size. What does this suggest about the order of P in $C_7(\mathbb{Q})$? (You are not asked to prove it.)

In Problem 2.5, parts (1)–(3) are the general theory and (5)–(7) the payoff for our family. If time is short, take (3) on faith and go to (5)–(7). Part (4) is the most satisfying: two completely different counts turn out to agree. Part (1) uses Bézout's theorem [Har77, §I.7].

Proposition 2.4 (The Hessian criterion)**.** *Let $C = V(F) \subset \mathbb{P}^2_{\overline{k}}$ be a smooth plane curve of degree m over a field of characteristic 0 or > 3, and let*

$$H(F) = \det\left(\frac{\partial^2 F}{\partial X_i\, \partial X_j}\right)_{0 \leqslant i,j \leqslant 2}.$$

Then a point $P \in C(\overline{k})$ is an inflection point if and only if $H(F)(P) = 0$. This is classical. Characteristic three is taken up in Problem 2.5(8) below.

Problem 2.5 (Inflection points and 3-torsion)**.**
(1) Let $C = V(F)$ be as in Proposition 2.4. Show that $H(F)$ is a form of degree $3(m-2)$, so of degree 3 when $m = 3$. Assuming C is a smooth plane cubic, explain why $C \not\subseteq V(H(F))$, and conclude from Bézout's theorem that C has exactly 9 inflection points *counted with multiplicity*.
(2) Now forget the Hessian. Let C be a smooth plane cubic over $\overline{k}$ and suppose $O \in C(\overline{k})$ is a flex, so that (C, O) is an elliptic curve with the group law of Problem 1.4(4). Show that

$$P \text{ is a flex} \iff [3]P = O,$$

so that the set of flexes is exactly $C[3](\overline{k})$.
(3) Show that a smooth plane cubic over $\overline{k}$ always *has* a flex, without using the Hessian.[a] Deduce that, with no flex singled out, the set of flexes is a torsor under $\mathrm{Pic}^0(C)[3]$. In particular the number of flexes is $\#C[3](\overline{k})$.

(4) Combine (2) and (3) with the structure of $C[3]$ in characteristic $\neq 3$ to conclude that a smooth plane cubic has exactly 9 flexes. What does comparing this with (1) tell you about the intersection $C \cap V(H(F))$?
(5) Specialize to C_d. Compute $H(F)$ for $F = X^3 + Y^3 - dZ^3$ and deduce that, as subsets of $C_d(\overline{k})$,

$$C_d[3](\overline{k}) = (C_d \cap \{XYZ = 0\})\,(\overline{k}).$$

Write down all nine flexes.
(6) Check (5) against the group law directly: use (2.4) to show that every point $P = (0 : Y_0 : Z_0) \in C_d(\overline{k})$ satisfies $[2]P = \ominus P$.
(7) Which of the nine flexes lie in $C_d(k)$? Treat the three lines separately, and say what happens for $k = \mathbb{Q}$.
(8) ($\star$) In characteristic three, $\#C[3](\overline{k})$ is 3 or 1 according as C is ordinary or supersingular, so a smooth plane cubic has 3 or 1 flexes. How is that compatible with (1)?

[a] *Hint:* for any $P_0 \in C(\overline{k})$ the class $\mathcal{O}_C(1) - 3[P_0]$ has degree 0, and multiplication by 3 is surjective on $\mathrm{Pic}^0 C$.

In Problem 2.8 you may use the following two results. Part (4) asks what the classification of automorphism groups in [Sil09, §III.10] predicts.

Theorem 2.6 (Morphisms fixing O [Sil09, Theorem III.4.8])**.** *A morphism of elliptic curves taking O to O is a group homomorphism.*

Proposition 2.7 (Endomorphism rings [Sil09, §III.4])**.** *The endomorphism ring of an elliptic curve has no zero divisors.*

Problem 2.8 (Complex multiplication, $\star$)**.**
Let $\alpha(X : Y : Z) = (X : Y : \zeta Z)$, a map defined over $k(\zeta)$.
(1) Show that α maps C_d to itself and fixes O.
(2) Conclude that α is a group homomorphism, hence an automorphism of the elliptic curve (C_d, O) over $k(\zeta)$.
(3) Show that $\alpha^2 + \alpha + 1 = 0$ in $\mathrm{End}(C_d \times_k k(\zeta))$, the ring of endomorphisms of C_d defined over $k(\zeta)$.[a]
(4) Deduce that $\mathbb{Z}[\zeta] \subseteq \mathrm{End}(C_d \times_k k(\zeta))$, so every Selmer curve has complex multiplication. What does the classification of automorphism groups then predict for $j(C_d)$ and for $\#\mathrm{Aut}(C_d)$ and $\#\mathrm{Aut}(C_d \times_k k(\zeta))$?

[a] *Hint:* $\alpha^3 = 1$, and the endomorphism ring of an elliptic curve has no zero divisors.

3. From Riemann–Roch to a Weierstrass equation

This section reproduces, for our family, the proof of [Sil09, Proposition III.3.1]: every elliptic curve is isomorphic to a curve in Weierstrass form.

Problem 3.4 is the longest problem here. Parts (1)–(3) go quickly once you see that Proposition 3.2 converts "pole order at O" into "multiplicity of O in $h \cap C_d$". Part (5) is a two-line computation with a surprising answer, and (8) is worth coming back to. Weierstrass equations and their coefficients $a_1, \ldots, a_6$ are as in [Sil09, §III.1].

Definition 3.1 (Riemann–Roch spaces)**.** For $D \in \mathrm{Div}\, C_d$,

$$\mathcal{L}(D) = \left\{ f \in k(C_d)^\times : \mathrm{div}(f) \geqslant -D \right\} \cup \{0\}, \qquad \mathrm{div}(f) \in \mathrm{Div}\, C_d.$$

This is a space of functions defined *over* k.

You may use the following two results without proof.

Proposition 3.2 (Bézout)**.** *For a nonzero linear form h, the scheme-theoretic intersection $h \cap C_d$ is an effective divisor of degree 3 in $\operatorname{Div} C_d$. Over $\overline{k}$ it becomes a sum of three points of $C_d(\overline{k})$ counted with multiplicity. Consequently, since $X + Y$ cuts out $3O$ by Problem 1.4(2),*

$$\operatorname{div}\left(\frac{h}{X+Y}\right) = (h \cap C_d) - 3O.$$

Proposition 3.3 (Riemann–Roch for genus one [Sil09, §II.5 and proof of Proposition III.3.1])**.** *For every $n \geqslant 1$ we have $\dim_k \mathcal{L}(nO) = n$. Since C_d is geometrically integral, $\mathcal{L}(D) \otimes_k \overline{k} = \mathcal{L}(D_{\overline{k}})$, so the dimension is the same whether computed over k or over $\overline{k}$.*

Problem 3.4 (The Weierstrass model of C_d)**.**
Consider the rational functions in $k(C_d)$

$$w \coloneqq \frac{Z}{X+Y}, \qquad v \coloneqq \frac{X-Y}{X+Y}.$$

For parts (2)–(6) and (8) assume $\operatorname{char} k \neq 2$. Part (7) replaces v by $X/(X+Y) = (1+v)/2$, which also makes sense in characteristic 2, and reaches a model valid whenever $\operatorname{char} k \neq 3$.

(1) Using Problem 1.4(1) and Proposition 3.2, compute $\operatorname{div}(w)$. Deduce that w has a pole of order exactly 2 at O and no other pole, so $w \in \mathcal{L}(2O)$. Where are its zeros, and what are they in the language of Problem 2.5?
(2) Show that the line $X - Y = 0$ does not pass through O. Deduce that v has a pole of order exactly 3 at O, so $v \in \mathcal{L}(3O)$. Where are its zeros?
(3) Using Proposition 3.3, explain why $\{1, w\}$ is a basis of $\mathcal{L}(2O)$ and $\{1, w, v\}$ is a basis of $\mathcal{L}(3O)$.
(4) The seven functions $1, w, v, w^2, wv, w^3, v^2$ all lie in $\mathcal{L}(6O)$. Why must they satisfy a non-trivial linear relation over k? List their pole orders at O. Which two collide, and why does that force the relation to involve both?
(5) Find the relation.[a]
(6) Let $E_1 \colon y_1^2 = x_1^3 - 432d^2$. Show that the relation of (5) says that there is a morphism $\varphi_1 \colon C_d \to E_1$ with $\varphi_1^* x_1 = 12dw$ and $\varphi_1^* y_1 = 36dv$. Write down φ_1 and its inverse explicitly.
(7) Where exactly did you use $\operatorname{char} k \neq 2$? Show that there is instead a morphism φ_2 with $\varphi_2^* x_2 = 3dw$ and $\varphi_2^* y_2 = 9dX/(X+Y)$ to

$$E_2 \colon y_2^2 - 9d\, y_2 = x_2^3 - 27d^2, \tag{3.1}$$

which is a Weierstrass equation valid whenever $\operatorname{char} k \neq 3$. Which of $a_1, \dots, a_6$ are nonzero?
(8) ($\star$) The relation in (5) has no w^2, wv, w or v term, although a general Weierstrass equation allows all four. Explain why, using the automorphisms $\sigma = \ominus$ of Problem 2.1 and α of Problem 2.8.[b]

[a] *Hint:* work on the affine chart $X + Y = 1$, where $X = (1+v)/2$ and $Y = (1-v)/2$, and expand $X^3 + Y^3$.
[b] *Hint:* compute $\sigma^* w$, $\sigma^* v$, $\alpha^* w$, $\alpha^* v$. Then use (4) to argue that the relation spans a one-dimensional space, so it is a simultaneous eigenvector.

4. Worked examples in Magma

Most of you have never opened `Magma`, so this section is written as worked examples rather than exercises: read them, then re-run them yourself and change things. Every line of output below is reproduced from `Magma` V2.28-8 [BCP97], with whitespace tidied and one long type name elided. Three short variations to try on your own are collected at the end.

Warning 4.1 (Closed points in `Magma`)**.** A closed point P of C_d is a point of the scheme C_d. Its residue field $k(P)$ is a finite extension of k, and $\deg P := [k(P) : k]$. The closed points of degree one are the rational points, and a closed point P with $k(P)/k$ separable corresponds to an orbit of $\deg P$ points of $C_d(\overline{k})$ under $\mathrm{Gal}(k^{\mathrm{sep}}/k)$. The divisor group $\mathrm{Div}\, C_d$ is the free abelian group on the closed points [Poo06, §1.9], whereas [Sil09, §II.3] works with points of $C_d(\overline{k})$ and Galois-invariant divisors on $\overline{C}_d$. `Magma` follows the first convention: `Places` returns closed points, `Divisor` builds elements of $\mathrm{Div}\, C_d$, and `RiemannRochSpace` returns $\mathcal{L}(D)$ as a k-vector space.

Each example continues the previous one. Begin with

```
QQ := Rationals();
P2<X,Y,Z> := ProjectiveSpace(QQ,2);
d := 7;
C := Curve(P2, X^3 + Y^3 - d*Z^3);
O := C![1,-1,0];
```

The notation `P2<X,Y,Z>` names the coordinates so that we can write equations in them, and `C!` coerces a list of three numbers into a rational point of `C`. Typing `O := (1:-1:0);` does not work.

Example 4.2 (Setting up, and a trap)**.** First, Problem 1.2:

```
IsNonSingular(C);  // true
Genus(C);          // 1
O;                 // (-1 : 1 : 0)
```

The last line is worth a pause. We asked for $(1 : -1 : 0)$ and got $(-1 : 1 : 0)$: a point of $\mathbb{P}^2(\mathbb{Q})$ is defined only up to scaling, and `Magma` normalizes the representative it stores, here by -1.

Now ask for the Weierstrass model. The second return value `mp` is the isomorphism from `C` to `E`.

```
E, mp := EllipticCurve(C, O);
E;
// Elliptic Curve defined by y^2 - 9/49*y = x^3 - 27/2401 over Rational Field
jInvariant(E);                          // 0
E eq EllipticCurve([0,0,0,0,-432*d^2]);   // false
```

That `false` looks alarming, since Problem 3.4(6) says the answer should be $E_1 : y_1^2 = x_1^3 - 432d^2$. It is not a contradiction: `eq` compares Weierstrass *coefficients*, and by [Sil09, Proposition III.3.1(b)] a Weierstrass model is unique only up to the changes of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$. The right question is whether the curves are isomorphic, and against which model:

```
Eours := EllipticCurve([0,0,-9*d,0,-27*d^2]);
Eours;                          // y^2 - 63*y = x^3 - 1323
bool, iso := IsIsomorphic(Eours, E);
bool;                           // true
IsomorphismData(iso);           // [ 0, 0, 0, 1/7 ]
```

So $r = s = t = 0$ and $u = 1/7 = 1/d$: `Magma` found exactly the model E_2 of Problem 3.4(7), with x_2 and y_2 divided by d^2 and d^3. The two maps agree as well,

```
DefiningEquations(mp);   // [ 3/7*Z, 9/49*X, X + Y ]
```

against our $(3dZ : 9dX : X + Y)$, the ratios being d^2 and d^3 exactly as $u = 1/d$ demands. The moral: for elliptic curves, `eq` is almost never the question you mean to ask.

We can even watch this happen for all d at once, by working over the function field $\mathbb{Q}(d)$ instead of over $\mathbb{Q}$:

```
Fd<D> := FunctionField(Rationals());
P2d<Xd,Yd,Zd> := ProjectiveSpace(Fd,2);
Cd := Curve(P2d, Xd^3 + Yd^3 - D*Zd^3);
Ed, mpd := EllipticCurve(Cd, Cd![1,-1,0]);
Ed;
// Elliptic Curve defined by y^2 - 9/D^2*y = x^3 - 27/D^4
DefiningEquations(mpd);
// [ 3/D*Zd, 9/D^2*Xd, Xd + Yd ]
```

which is (3.1) rescaled by $u = d$, as an identity in d.

Finally, the predictions of Problem 2.8(4):

```
#AutomorphismGroup(E);                    // 2
K<zeta> := CyclotomicField(3);
#AutomorphismGroup(BaseChange(E,K));      // 6
```

Example 4.3 (The addition formulas, checked)**.** A `Magma function` takes arguments and returns a value. Here both are sequences of three coordinates, so that (2.3) and (2.4) can be typed almost verbatim.

```
dbl := function(P)
  x := P[1]; y := P[2]; z := P[3];
  return [-y*(2*x^3+y^3), x*(x^3+2*y^3), z*(x^3-y^3)];
end function;

add := function(P1,P2)
  A := P1[1]^2*P2[1] + P1[2]^2*P2[2] - d*P1[3]^2*P2[3];
  B := P1[1]*P2[1]^2 + P1[2]*P2[2]^2 - d*P1[3]*P2[3]^2;
  return [B*P1[2]-A*P2[2], B*P1[1]-A*P2[1], B*P1[3]-A*P2[3]];
end function;

P := [2,-1,1];
Q := dbl(P);      C!Q;   //  (5/3 : 4/3 : 1)
R := dbl(Q);      C!R;   //  (-1256/183 : 1265/183 : 1)
S := add(P,Q);    C!S;   //  (-17/38 : 73/38 : 1)
```

These are the rational points of Problem 2.3. Coercing through `C!` normalizes them, which is why $(15 : 12 : 9)$ prints as $(5/3 : 4/3 : 1)$. Now transport everything to E through `mp` and compare with `Magma`'s own group law:

```
EP := mp(C!P);
mp(C!Q) eq 2*EP;          // true
mp(C!S) eq 3*EP;          // true
mp(C!R) eq 4*EP;          // true
```

Three `true`s: formulas derived from nothing but (2.1) and a 3×3 determinant agree with a general-purpose implementation.

Example 4.4 (Riemann–Roch, mechanically)**.** `Magma` can carry out Problem 3.4 without being told the answer. The generators `x` and `y` of `FunctionField(C)` are the affine coordinates X/Z and Y/Z of C_d (not Weierstrass coordinates), so our two functions are built from those.

```
FF<x,y> := FunctionField(C);   // x = X/Z,  y = Y/Z
w := 1/(x+y);                  // Z/(X+Y)
v := (x-y)/(x+y);              // (X-Y)/(X+Y)
pl0 := Place(0);
Valuation(w, pl0);             // -2
Valuation(v, pl0);             // -3
```

Those are the pole orders of Problem 3.4(1), (2). A valuation of -2 is a pole of order 2. As in Warning 4.1, `Place(0)` is the closed point O and `Divisor` builds an element of $\operatorname{Div} C_d$, so everything here happens over $\mathbb{Q}$. Next, Proposition 3.3 itself:

```
D := Divisor(pl0);
for n in [1..6] do print n, Dimension(RiemannRochSpace(n*D)); end for;
// 1 1
// 2 2
// 3 3
// 4 4
// 5 5
// 6 6
```

Now ask for a basis of $\mathcal{L}(2O)$ without suggesting one:

```
V2, m2 := RiemannRochSpace(2*D);
[m2(b) : b in Basis(V2)];      // [ x^2 - y*x + y^2, 1 ]
```

which looks nothing like our w. But $X^3 + Y^3 = (X+Y)(X^2 - XY + Y^2)$, and on C_d the left side is dZ^3, so

$$\frac{X^2 - XY + Y^2}{Z^2} = \frac{dZ^3}{Z^2(X+Y)} = d\,\frac{Z}{X+Y} = d\,w,$$

and indeed

```
(x^2 - x*y + y^2) eq d*w;      // true
```

Finally the relation. We list the seven functions of Problem 3.4(4), express each in a basis of $\mathcal{L}(6O)$ (that is what `@@` does, applying the inverse of the map `mV`), and take the kernel of the resulting matrix.

```
V, mV := RiemannRochSpace(6*D);
fns := [FF!1, w, v, w^2, w*v, w^3, v^2];
[-Valuation(f, pl0) : f in fns];   // [ 0, 2, 3, 4, 5, 6, 6 ]
M := Matrix([Eltseq(f @@ mV) : f in fns]);
K := Kernel(M);
Dimension(K);                      // 1
Basis(K);                          // [ ( 1  0  0  0  0  -28  3) ]
```

The pole orders are $0, 2, 3, 4, 5, 6, 6$: seven functions, six available orders, the collision at 6 forcing the relation exactly as in Problem 3.4(4). And the kernel is one-dimensional, spanned by the vector reading

$$1 - 28w^3 + 3v^2 = 0,$$

which is $3v^2 + 1 = 4dw^3$ at $d = 7$. We got it by hand from a factorization. `Magma` got it from linear algebra in a Riemann–Roch space.

Now try these. Each one is a small edit to Example 4.2 or Example 4.3, and in each case the worksheet already tells you what the answer must be.

(1) Set `d := 2`. Problem 2.1(3) predicts a rational point of order two. Find it, verify with the group law that it has order two, and compute its image under φ_1 by hand.

(2) Set `d := 8`. Problem 2.5(7) predicts rational 3-torsion. Find the two rational 3-torsion points other than O, use your `dbl` function to check that $[2]P = \ominus P$, and compute $\varphi_1(P)$ by hand.

(3) Use `IsIsomorphic` to decide whether $C_1 \simeq C_8$ and whether $C_1 \simeq C_2$ over $\mathbb{Q}$. Pass it the *Weierstrass* models: on the plane curves themselves `Magma` answers `For genus 0 or 1 curves the basefield must be finite`. Reconcile the answers with the cubic-twist criterion of Section 5.

5. Where the name comes from

Selmer [Sel51] studied the general diagonal cubic $aX^3 + bY^3 + cZ^3 = 0$, and produced the standard counterexample to the Hasse principle: the curve $3X^3 + 4Y^3 + 5Z^3 = 0$ has real points and $\mathbb{Q}_p$-points for every prime p, but no rational point. Such a curve is a torsor under its Jacobian [Poo17, §5.12], and that Jacobian is a Selmer curve in our sense:

$$\mathrm{Jac}(aX^3 + bY^3 + cZ^3) \simeq \big(y^2 = x^3 - 432(abc)^2\big).$$

Two ways to test this in `Magma`: the curve $X^3 + 2Y^3 + 3Z^3 = 0$ *does* have the rational point $(1 : 1 : -1)$, so it is its own Jacobian and should be isomorphic to C_6, that is to $y^2 = x^3 - 432 \cdot 6^2$; and for Selmer's own curve one can call `Jacobian(GenusOneModel(3*X^3 + 4*Y^3 + 5*Z^3))` and compare with $y^2 = x^3 - 432 \cdot 60^2$. Both come out `true`.

Finally, by Problem 3.4(6) together with the classification of twists of $y^2 = x^3 + A$ [Sil09, §X.5], for $\mathrm{char}\, k \neq 2, 3$ the curves C_d and $C_{d'}$ are isomorphic over k exactly when $d/d' \in (k^\times)^3$, so the family is parametrized by $k^\times/(k^\times)^3$. These are the cubic twists of $X^3 + Y^3 = Z^3$, and they are the natural first examples for the descent machinery of [Sil09, §X] and [Poo17, Ch. 8].

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